Introduction to Mathematics and Optimization

May 2025

The exam answers must be based on the syllabus. Concepts and results in the answer that are not part of the syllabus must be explained using the syllabus.

PROBLEM 1

Let C denote the subset of $\mathbb{R}^2$ given by

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 3,\ y \leq 3,\ x + y \geq 4\}$$

and let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function given by

$$f(x, y) = 3x^2 + 2y^2.$$

Consider the optimization problem

$$\begin{array}{cc} \min & f(x,y) \\ (x,y) \in C. & \end{array} \tag{1}$$

(a) State the interior C^o and the boundary ∂C of C.

(b) Show that C is compact and that (1) has an optimal solution.

(c) Show that if $(x, y) \in C$ is an optimal solution to (1), then $(x, y) \in \partial C$.

(d) Show that (1) is a convex optimization problem with a unique solution.

(e) Write down the Karush-Kuhn-Tucker conditions for (1).

(f) Solve the optimization problem (1) using a method from the syllabus that is based on $C = C^o \cup \partial C$.

PROBLEM 2

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be the function given by

$$f(x, y, z) = x^2 + xy + y^2 + xz + z^2 + yz$$

for $(x, y, z) \in \mathbb{R}^3$.

(a) Find a symmetric 3×3 matrix A such that

$$f(x, y, z) = \begin{pmatrix} x & y & z \end{pmatrix} A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

for all $(x, y, z) \in \mathbb{R}^3$.

(b) Show that A is positive semidefinite.

(c) Show that $f(x, y, z) \geq 0$ for all $(x, y, z) \in \mathbb{R}^3$.

(d) Find all $(x, y, z) \in \mathbb{R}^3$ such that $f(x, y, z) = 0$.

(e) Is f a strictly convex function?

(f) Show that $(0, 0, \frac{1}{2})$ is a solution to the optimization problem

$$\begin{gathered} \min f(x, y, z) \\ x + y + 2z = 1 \end{gathered}$$

by using the KKT conditions as they are stated in the syllabus. Do the KKT conditions have a unique solution?

PROBLEM 3

Let $C \subseteq \mathbb{R}^3$ be the set of points $(x, y, z) \in \mathbb{R}^3$ that satisfy the following inequalities:

$$\begin{aligned} x + y - z &\leq 1 \\ -x + 2y + z &\leq 2 \\ x - y + z &\leq 3 \\ -x - y - z &\leq 4. \end{aligned}$$

(a) Use Fourier-Motzkin elimination to eliminate z from the inequalities.

(b) Next eliminate y from the inequalities you obtained in (a).

(c) Show from (b) and (a) that C is bounded.

(d) Show that C is compact.

(e) Let the function $f : \mathbb{R}^3 \to \mathbb{R}$ be given by

$$f(x, y, z) = 12 \cos(\sin(e^{x^2 - y^3}) + y^{17}) + 15xy - e^{x^2 - 5x + 6}.$$

Does the optimization problem

$$\begin{aligned} &\min f(x, y, z) \\ &(x, y, z) \in C. \end{aligned}$$

have a solution? Justify your answer.